ICU Bed Occupancy Change Prediction

Final Project Report for Project Group PG20

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***Abstract*—Reliable short-term ICU occupancy forecasting has become increasingly important as hospitals face fluctuating admission demands and rising pressure on critical care capacity. The primary objective of the project is to develop a hybrid ICU occupancy bed prediction model that integrates new admission dynamics and the expected LoS for existing patients. The model focuses on predicting short-term ICU occupancy by combining projected new ICU admissions and the remaining length of stay of current patients, by using the MIMIC-IV dataset. The hybrid framework combines the admission prediction model, which estimates daily or cumulative inflow through a rolling-origin forecast, with the LoS prediction model, which provides patient-day–level remaining-stay estimates to characterize expected outflow. Across all horizons, the hybrid model captures the overall trend of ICU occupancy, and short-term accuracy is reasonable. Occupancy-change predictions remain relatively stable across horizons, suggesting consistent ability to track short-term increases or decreases. The hybrid model successfully characterizes both inflow and outflow dynamics over short planning windows. Overall, the hybrid model offers a more realistic representation of ICU flow and supports operational decision-making in short-term ICU bed management.**

I. Introduction

A. Context

Hospitals often struggle to provide timely ICU access for critically ill patients, while unused beds lead to unnecessary operating costs. Many existing forecasting practices overlook the combined effects of new admissions and patient discharges, and simple average-based LoS assumptions can distort capacity estimates and weaken resource planning. Because patient LoS shifts across different periods, relying on fixed averages frequently underestimates actual occupancy and highlights the need for more accurate modelling. ICU's occupancy forecasting is a highly consequential task, and the COVID-19 pandemic exposed how vulnerable ICU capacity can be without reliable, adaptable prediction systems. Accurate forecasts help hospitals optimize staffing, reduce unnecessary diversions, and maintain care quality under uncertainty.

B. Use Case

1) Business Context: This project addresses the practical challenge of managing short-term ICU bed availability, where hospitals must balance keeping sufficient reserve capacity with avoiding empty beds. During periods such as the COVID-19 pandemic, ICU occupancy in some regions rose above 90%, underscoring how important timely, data-supported planning can be. Having a reasonable forecast of near-term occupancy helps hospitals make clearer decisions about staffing, scheduling, and the use of critical resources.

2) Key Stakeholders/Users: The forecasts are mainly intended for ICU operations managers and hospital administrators. They can also support teams responsible for clinical coordination, as well as information systems involved in assigning beds and managing workload. While the system is designed with individual hospitals in mind, it can be adapted for network-level use if the infrastructure allows.

3) Target Variable: The model focuses on predicting short-term ICU occupancy. This estimate is obtained by combining two elements: projected new ICU admissions and the remaining length of stay of current patients. These components are modeled separately and then merged within a hybrid structure to represent how beds are used and freed over time.

4) Time Window: The forecasting horizon spans 3 to 7 days, which fits the planning cycle typically used for ICU admission control and resource allocation. Although unexpected emergency cases are more difficult to anticipate, the model is updated every day so that new admissions can be incorporated and future estimates adjusted accordingly.

C. Problem Statement

Hospitals face challenges in providing ample beds in ICU for critically ill patients in a timely manner, while underutilized beds increase the cost for hospitals. Most current operational practices do not consider the interaction of admissions and discharges based on the LoS in forecasting the ICU bed occupancy rate, and LoS is sometimes modelled naively or simply as the average value. This leads to overly optimistic ICU capacity forecasts, thereby compromising bed management and resource allocation decisions. Patient LoS varies across different phases, and using an average LoS underestimated true bed occupancy, indicating the importance of modelling the LoS accurately. Inaccurate modelling of the LoS will underestimate bed occupancy rate prediction and reduce the hospital administration's preparedness in providing more beds or reducing the stay of non-critical patients. During the Covid 19 pandemic, the increase in bed occupancy due to the unpredictability of LoS greatly contributed to the mortality rate.

D. Objectives

The primary objective of the project is to develop a hybrid ICU occupancy bed prediction model that integrates new admission dynamics and the expected LoS for existing patients. LoS forecasts will leverage individual-level data to provide more accurate and widely applicable estimates of each patient's stay, while addressing the systematic underestimation and lack of adaptability of naïve methods that rely solely on historical averages. This model aims to improve bed occupancy rate prediction over a 3–7-day horizon, achieving higher accuracy and reducing mean errors compared to admission-only or LoS-only frameworks. Beyond that, the model's outputs are expected to be translated into practical decision-support tools for hospital administrators, with an early warning mechanism to alert

administrators of potential under- or over-occupancy and support timely bed management.

E. Assumptions

We assume that the MIMIC-IV dataset provides a reasonable reflection of ICU operations and that missing values follow MAR. ICU capacity is treated as stable for the next three to seven days. The model focuses mainly on the remaining LoS for patients who are already admitted, which means very short new admissions during the window may not be fully represented. Sudden surges or major policy changes are not included within the scope of the forecast.

II. Related Work

Research on ICU bed occupancy rate prediction could follow three main approaches: directly forecasting occupancy levels, modeling patient admissions and predicting length of stay (LoS). For direct occupancy forecasting, models such as Support Vector Regression, eXtreme Gradient Boosting, Random Forest, K-Nearest Neighbors, Gradient Boosting and Decision Tree have been applied, and these machine learning methods usually prevail over traditional methods such as ARIMA in capturing complex trends and multivariate dependencies. Deep learning has also shown promise, with the NARX recurrent neural network outperforming the naive baseline in most MAE evaluations. In admission modeling, hybrid approaches that combine SEIR equations and neural networks have been shown by Riccardo Delli Compagni et al. (2022) to outperform SEIR alone across hospitals of different sizes. For LoS prediction, Momo et al. (2024) demonstrated that LSTM models can capture heterogeneity across units, age groups and disease groups, and domain adaptation further improves accuracy and computation time.

A comparative analysis of eight selected studies shows that data-driven models consistently outperform traditional forecasting approaches for short-term predictions. Several studies highlight the importance of LoS and note that ignoring censored data leads to significant underestimation. However, differences in data scope and model complexity remain, with some studies relying on limited datasets and others integrating multimodal data. Many models are validated only in narrow contexts, censored LoS data are often neglected, and external factors are rarely incorporated quantitatively. Medium-term forecasting horizons remain underexplored despite their relevance to ICU planning.

Our project addresses these gaps by proposing a hybrid model that explicitly combines admission dynamics with patient-level LoS prediction. This approach moves beyond admission-only or LoS-only frameworks and incorporates rich individual-level data to improve accuracy, robustness and real-world decision support.

III. Methodology

A. Workflow Overview and Step-by-Step Process

The workflow used in our project is shown in Figure 1. During the data preparation phase, we first selected publicly-available datasets which are suitable for our project objective and then integrated the data we needed, before performing and data preprocessing and data quality check. Through EDA, we gained more insights into data structures and its key patterns, which guided us to make further adjustments to the problem settings and use cases.

Then, we constructed the hybrid model framework, which comprised two major independent but integrable modules – LoS predictions and admission forecasts. Prior to model construction, the cleaned datasets were split into train and test test based on same standard, so as to ensure consistency in subsequent model training and evaluation.

Subsequently, training, tunning and evaluations of the models in the two modules were conducted parallelly to make full use of time and each team member's strengths. After repeatedly optimizing the model structure and performances, we finally integrated the outputs of two sub-models according to the hybrid model structure to generate the final predictions and overall evaluations. We kept refining the integration strategies while handling issues such as missing values and date alignment during the process.

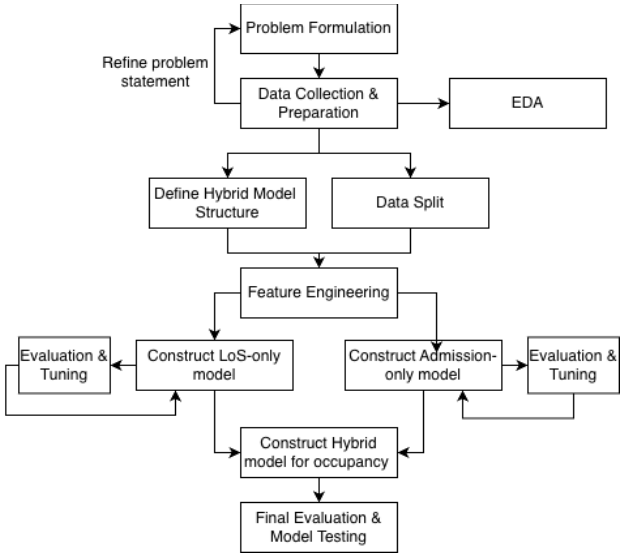

Figure 1. Overall Workflow Summarizing the Key Phases of the Project

B. Code Repository and Replication Instructions

Our source code is organized in Jupyter notebooks. The `Dateset/` folder stores raw data while `Merged_Dataset/`is used to store the datasets after merging, preprocessing as well as feature engineering, which are the data directly used for model evaluation. The `Predicition/` folder includes predictions generated from LoS-only models, admission-only models and also the final hybrid models. The ensure the reproducibility of the experiment, all the dependencies are listed in the `requirement.txt` file for environment setsup.

Prior to running any code in the repository, downloading MIMIC-IV dataset and placing it in `Dateset/`are required. Then the notebook files can be executed as the following order:

- `DataMerging.ipynb`: Select and merges multiple data tables provided in MIMIC-IV.
- `Preprocessing.ipynb`: Perform data cleaning and preprocessing and add more features to support further analysis.
- `Create_Embedding.ipynb` and `Training.ipynb`: Generate embeddings and traing LoS prediction models.

- `Admission_forecast_attempts.ipynb`: Construct and evaluate the admission forecasting models.
- `Hyrbid Model for ICU Occupancy Prediction.ipynb` and `Test_Set_Imputation.ipynb`: Integrate outputs generated from LoS and admission predictions and perform the final evaluation.

IV. Data Collection and Pre-processing

A. Data Collection

The dataset used is the MIMIC IV[1], a large deidentified EHR dataset of patients admitted to the emergency department or an intensive care unit at the Beth Israel Deaconess Medical Center in Boston, MA. MIMIC-IV contains data for over 65,000 patients admitted to an ICU, and over 200,000 patients admitted to the emergency department spanning 11 years from 2008 to 2019. As the MIMIC IV website pointed that - "Importantly, data cleaning steps were not performed, to ensure the data reflects a real-world clinical dataset.", this posed a challenge in data quality especially in handling lab measurements where they are prone to errors. Notwithstanding these issues, the dataset provides massive data of patients available upon admission and time series data of the medications, vital sign measurements and procedures taken as the patients stay in the ICU.

B. Data Variables

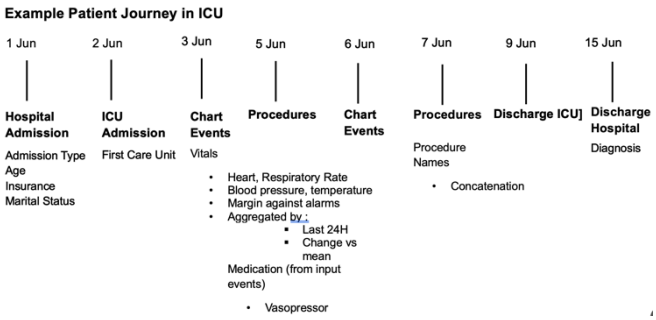

Figure 2. Example of a patient's journey in ICU

The features can be logically grouped into 4 main categories (see Appendix A):

Hospital admission data: Admission data captured once at the start of the stay in the hospital

ICU admission data: Admission data captured once at the start of the stay in the ICU

Chart events: Time-series data reflecting the patient's real-time physiological state, captured as multiple events in a day. This is joined with the input events table which contains medication taken.

Procedure: The procedures taken by the patient during ICU

C. Exploratory Data Analysis (EDA)

This section summarizes the key findings from the EDA of our dataset (see Appendix C for all figures and detailed visualizations). The analysis focuses on data quality, distributional characteristics, structural patterns, temporal behaviors, and inter-variable relationships relevant to downstream ICU occupancy modeling.

Data Quality and Structure

Automated profiling (YData, Sweetviz, DataPrep) showed no missing values, no duplicated rows, and consistent variable types, indicating strong data integrity. Several pairs of variables – such as blood pressure measurements, ICU unit identifiers, and liver-disease indicators – were found to be highly correlated or redundant (Fig. A1), which informed later feature consolidation. A small number of negative *stay_days* values resulted from MIMIC-IV timestamp shifts, but no logically invalid records were observed. These anomalies were documented and addressed during preprocessing.

Distributions, Patterns, and Trends

Most numeric variables exhibit right-skewed or long-tailed distributions, typical of ICU utilization (e.g., *stay lengths*, *chart event counts*). Demographic variables show expected concentration in older adults, males, and English-speaking patients. Intervention-related features are sparse with many zero values, while length-of-stay variables follow an exponential-decay pattern—short stays are common, but long stays have a disproportionate impact on occupancy. Temporal variables are uniformly distributed across anonymized years (2110–2214), with no seasonal or structural irregularities, confirming stable temporal coverage for forecasting.

Variable Relationships

Correlation and association analyses highlight both redundant encodings (e.g., *first_careunit* vs. *icu_types*, *discharge_location* vs. *hospital_expire_flag)* and clinically meaningful dependencies (e.g., strong association between *mld* and *msld*). These findings guided the removal or consolidation of overlapping features and helped prioritize variables for modeling.

Key Insights for Modeling

- The dataset is clean and structurally consistent, requiring minimal preprocessing beyond handling timestamp anomalies.
- Redundant and highly correlated features should be combined to avoid multicollinearity.
- Heavy skewness, sparsity, and imbalance imply that robust modeling techniques (e.g., regularization, class-imbalance handling, log-transformations) may be beneficial.
- Stable temporal patterns confirm that time-dependent forecasting models can be applied without seasonality adjustments.

These insights informed feature selection, data preprocessing, and the modeling strategies developed in subsequent sections.

D. Data Pre-processing

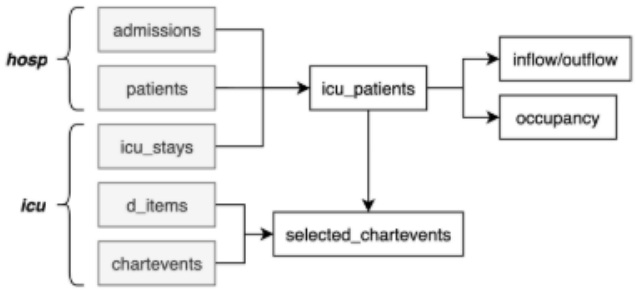

Figure 3. Source datasets used to produce selected_chartevents as the main dataset

ICU patients' profiles

The icu_patients table incorporated ICU patients' admission and discharge information (e.g., time, location, reasons) as well as their demographics (e.g., gender, race, insurance status), by combining admissions, patients and icu_stays. To address inconsistencies caused by manual data entry, such as variations in ICU department names, basic standardization procedures were applied.

Subsequently, daily inflows and outflows were computed based on the admission and discharge time of ICU patients after tackling the shifted year issues to explicitly show the ICU dynamics over time.

Clinical records from chartevents

To better measure the patients' conditions and predict their LoS, clinical records during ICU stays are very crucial. Due to the large number of chart event types and sparsity of measurement records, feature selection is necessary before extraction.

The raw chartevents data were joined with d_items first to obtain descriptions for each charted examination or screening. Text-based items were excluded at this stage to simplify data processing. The events were then selected mainly based on the frequency counts of each type (i.e., labels in the original data), while certain events were specifically preserved as they potentially indicate high patient severity.

Additionally, several variables with similar meaning but different labeling (e.g. height, weight, temperature) were identified. The selected items were checked one by one to detect such issues and standardized by assigning the same labels. Measurement units for these records were also converted to ensure consistency. Finally, all charted records were aggregated daily, to reduce the data scale and align with the project requirements.

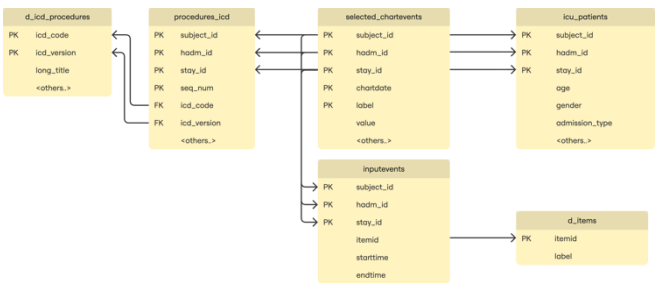

Figure 4. Training dataset produced by joining the main dataset – selected_chartevents with related tables to incorporate additional features

The datasets were merged according to the ER diagram shown in Figure 1. Data quality is assessed across several factors: structure, missing values and outliers.

i. Data Cleaning

The columns were validated to have consistent data types : strings, numeric and dates. The *selected_chartevents* contained selected patient's lab measurements. Each measurement corresponds to a specific measurement, date and time. In order for the chart events to be used as features for machine learning, these measurements were pivoted into columns. Each row now represents the measurement for a specific patient under a specific hospital admission and ICU stay at a specific chart date. This was a time series data collected at irregular intervals. For LoS predictions, a machine learning approach was used, in view of the many features such as patient demographics and admission details that are not time-based but beneficial for prediction.

The lab measurements were then aggregated in a time window from ICU admission to the current chart date and the target has to be modified to the remaining LoS, and not the entire LoS to prevent leakage as shown in Figure 9.

ii. Missing Values

Missing values primarily occur in the lab measurements after pivot, being not consistent across all patients. Some measurements such as heart rate and arterial blood pressure were more taken more frequently than the rest. However, this is considered Missing At Random because not having a test taken could mean that the patient is healthy with respect to the test. The strategy for dealing with missing value differ between categorical and numerical features. Categorical features in the dataset were the binary events for a patient such as if a patient has undergone dialysis. Handling missing value was done by assigning a new category. Numerical features were handled by the multivariate imputation by chained equations MICE[17] algorithm. Instead of filling in a single value for each missing data point such as the mean, which is the common method, the algorithm iteratively predicts the missing value by observing the other variables in the dataset.

iii. Outliers

Outliers appeared in the lab measurements as well, and not in the other features. Some of the values were beyond what is physiologically normal for a human being. For example, heart rates of 10000 and weight of 14000kg were found in the values. At this stage, the outliers based only on what is physiologically impossible were categorized rather loosely based on research on the web due to lack of medical knowledge. For example, it's difficult to determine the maximum weight a human can have other than searching for who is/was the heaviest person alive. Additional outlier detection was enforced in the model training stages. Furthermore, some algorithms such as decision trees are less sensitive to outliers.

After the outliers were categorized, they were assigned as missing values before being treated using the same MICE algorithm described in the previous section.

The target is severely right-skewed and is clipped at 21 days (21 days generally accepted for long stay patients [19])

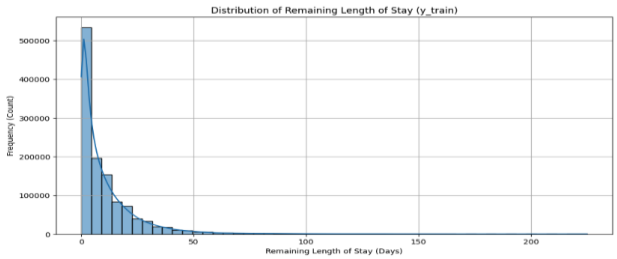

Figure 5. Target distribution

iv. Feature selection

A correlation map (Figure A1) was created to look at the predictive power of the features - dialysis, invasive lines and vasopressors showed higher correlation with the target which made sense because they are indicators of severely ill patients.

There were some multicollinearities detected in the list of features between the last 24 hours and change values for a specific vital sign as they are related. Since the models used later were trees-based models that can handle multicollinearities inherently, the features were allowed but taken into account during the SHAP importance interpretation.

Another feature that was dropped was stay_days – the number of days that has transpired since the patient was admitted to the ICU. This feature could potentially introduce survivorship bias to prevent the model from learning useful features that affect patients to stay longer in the ICU other than the obvious fact that they were already critically ill.

Admission type, admission location, first care unit and ICU types are also tightly correlated, and so only the first care unit was included as a feature.

v. Scaling and encoding

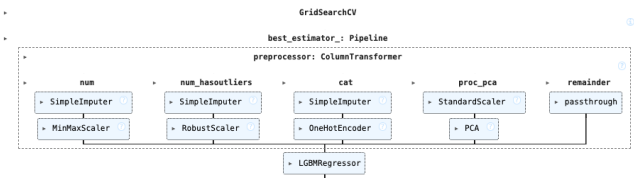

Figure 6. Features preprocessing pipeline

Sklearn's pipeline was used to perform the feature preprocessing steps for different features. For numerical features that have outliers (vital signs), RobustScaler that scales the value based on IQR range was used. For other numerical features where outliers were not found (such as age) or sufficiently dealt with earlier (such as binary vital signs), the MinMaxScaler was used instead. One hot encoding was used for the categorical features. Text embeddings were used to encode the procedure names using BERT embeddings (explained further in feature engineering). The embeddings have 768 dimensions and PCA was used to reduce the number of features to avoid overfitting the model. The first 50 components of PCA were found to explain 90% of the variance.

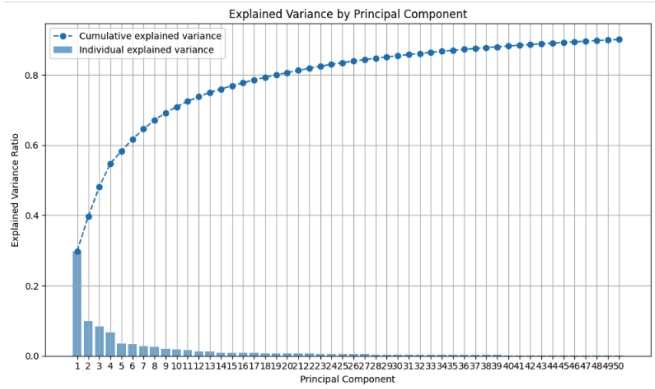

Figure 7. PCA explained variance

V. Core Concept, Methods, and Models

A. Feature Engineering

These are the list of additional features that were engineered:

i) *Last 24H <vital sign>*: Represents the average in the last 24 hours. (Table 3)

ii) For numerical vital sign measurements,

Change in <vital sign>: $\frac{v_{\text{last_24h}} - \bar{v}_{\text{daily_mean}}}{\bar{v}_{\text{daily_mean}}}$
where v is the vital sign (Table 3)

iii) For binary vital sign measurements,

Change in <vital sign>: $v_{\text{last_24h}} - \bar{v}_{\text{daily_mean}}$

where v is the vital sign, to avoid divide by zero scenarios (Table 3)

iv) For heart rate and respiratory rate, the difference with the alarms highs and lows were calculated in order that the model could learn the physiological values that are clinically important (Table 3)

v) The number of vasopressor drugs taken was calculated to represent critically ill patients especially those who are in shock as the last 24 hours and change from the mean values (Table 3)

vi) The concatenated list of procedures was computed by checking the procedure date against the chart date to ensure that there is no leakage. Since this is a textual feature, two approaches were taken: Tf Idf and embeddings. The embeddings approach, based on BioClinicalBERT[18] resulted in a lower 0.5 in MAE. The BioClinicalBERT embeddings is the ClinicalBERT (BERT model pre-trained on a corpus of biomedical text) that was further pre-trained on MIMIC notes, available in HuggingFace[18] as both model and tokenizer. The higher accuracy could be attributed to the ability of the embeddings to encode related concepts as compared to simply mapping the frequent terms in Tf-Idf. The embeddings (440,000 rows) were computed using Pytorch on a local Mac Studio (MPS architecture) which took 20 minutes.

B. Model Construction

To obtain a more accurate short-term ICU occupancy prediction, we combine the admission prediction model (which estimates daily/cumulative inflow) and the LoS prediction model (which estimates each patient's remaining stay duration).

The hybrid model is formulated as follows:

$O_{t+d} = N_t + C(d) - \sum_{i=1}^{N_t} 1(L_{i,t} \leq d)$ where

O_{t+d} :predicted ICU occupancy at day t+d

N_t: current ICU occupancy (number of patients currently admitted)

$C(d)$: cumulative predicted number of new admissions across all d days (from the admission model)

$L_{i,t}$: predicted reamining length of stay for patient i at time t (from the LoS model)

$1(L_{i,t} \leq d)$: indicator function representing whether patient i is expected to be discharged within the next d days

The occupancy change: $\Delta O_{t+d} = O_{t+d} - O_t$

This hybrid formulation was chosen because neither admission forecasting nor LoS modelling alone can fully capture ICU occupancy dynamics. The admission model provides accurate forward-looking inflow, while the LoS model offers individualized discharge expectations. By structurally combining them, the hybrid model aligns with real-world operational logic: occupancy evolves from today's census plus expected arrivals minus expected departures. Moreover, the explicit use of patient-level LoS predictions enables the model to incorporate heterogeneity in disease course and care intensity-an advantage over aggregate discharge-rate models that assume homogeneous service times.

C. LoS Prediction

The objective of the LoS model is to predict the remaining length of stay of patients in ICU during admission and throughout the ICU stay

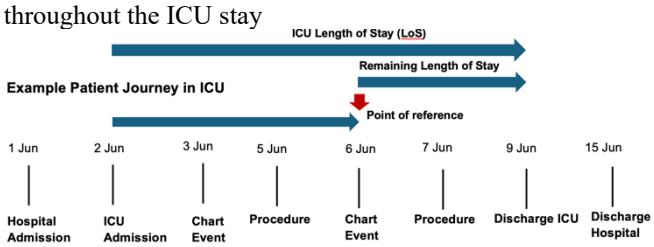

Figure 9. Remaining length of stay derivation

In order to prevent leakage, a time-based split was performed on an 80% train and 20% test ratio. An initial GridSearch CV based on default parameters was used to include the Sklearn pipeline created for the feature preprocessing on a cv of 3 on 3 algorithms – Ridge Regression, XGBoost and LightGBM for model selection. Ridge regression was used to serve as a baseline model for sanity check with L2 regularization to prevent overfitting since some multicollinearities were detected earlier. XGBoost is a powerful tree-bases gradient boosting algorithm that can capture complex non-linear relationships in the data; and LightGBM is often used to compare with XGBoost being a more efficient tree-building technique. As such, LightGBM can prove to be viable due to computational effectiveness if the accuracy is close to XGBoost. The results are shown below:

	Ridge Regression	XGBoost	LightGBM
MAE	5.28	4.97	4.96
RMSE	6.49	6.65	6.66

Table 5: Initial model results for model selection

Due to outlier in the target, the MAE being less sensitive to outlier was considered. Based on the results, LightGBM was selected and the hyperparameters were tuned by first selecting a high n_estimators (1000) and enabling early stopping. A value of 651 boosting iterations was completed before stopping. Next, the 651 was set as the n_estimators and a grid of parameters (learning rate, number of leaves and lambda) for cv=5 were searched for the optimal accuracy. The results after hyperparameter tuning are shown below with n_estimators=651, learning_rate = 0.05, num_leaves = 31 and reg_lambda = 3:

	LightGBM
MAE	4.93
RMSE	6.63

Table 6: Results after hyperparameter tuning

D. Admission Forecasts

The objective of the admission forecast phase is to estimate the cumulative number of new admissions over the next d days, while d is the forecast horizon specified before. Instead of directly predicting the total admissions following end-to-end pipeline, a two-step approach is adopted: first, we predict the daily admissions over the next d days in a rolling-origin manner; then, the daily forecasts are summed up to obtain the total admissions within the horizon.

Technically, the steps can be interpretated as follows. At each time point t, the model generates the daily admissions $\hat{A}_{t+k}, k = 1, \dots, d$, then the total sum would be $\sum_{k=1}^{d} \hat{A}_{t+k}$, where:

$$\hat{A}_{t+k|t} = \mathbb{E}(A_{t+k} | A_1, \dots, A_t)$$

Such approach can better capture the daily variations of the ICU new admissions, while enhancing interpretability and flexibility of model outputs. The overall forecast horizon for the hybrid model can be easily adjusted by simply modifying the summation window, thereby reducing repeated computational costs and better aligning with actual decision-making flows.

As shown in Figure 2, the daily admissions (based on the re-assigned years) maintain a relatively stable level even though despite fluctuations before 2023. After 2023, however, the inflow number plummet with the average daily admission dropping from 17 to around 3.5.

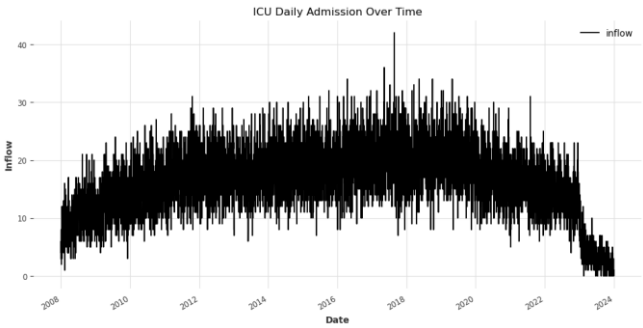

Figure 2. Trend of ICU daily admissions

The STL decomposition was subsequently applied prior to model selection in order to assess the strength of trend and seasonality. The resulting values indicate there exists a significant long-term trend with strength at 0.685, with only the moderate seasonality patterns at 0.325. Therefore, we selected exponential smoothing models (including Holt's Linear Model and Holt-Winters Seasonal models) as well as ARIMA-type models (e.g. SARIMA) as our primary choices, together with several baseline models, such as the naïve mean and moving average models.

During the process, new features like current bed occupancy are also considered to be included. However, after checking the correlation between these variables and the admissions, the statistical relationship is quite weak, and would only lead to overfitting rather than improvement. Also, since the dates are shifted and cannot reflects the true scenarios, the weekday number features would also be not reliable. So, the prediction solely relies on its own patterns.

VI. Evaluation of Results and Findings

A. LoS

1) Model Performance for short stay and long stay patients

The results after hyperparameter tuning only showed marginal improvement as depicted in training progress below where training tapered down after 100 boosting iterations which is LightGBM default n_estimators.

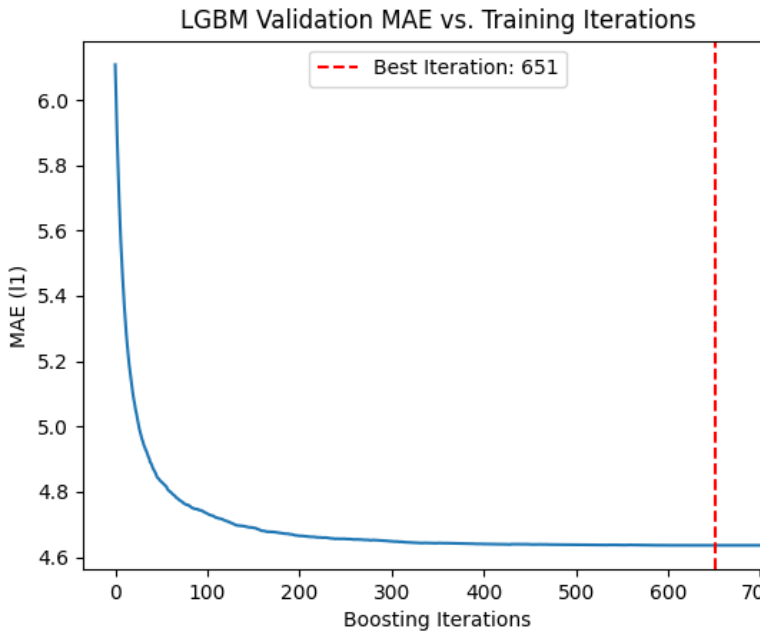

Figure 10: Training progression

Upon further investigation, the accuracy was better for short stay patients as shown below:

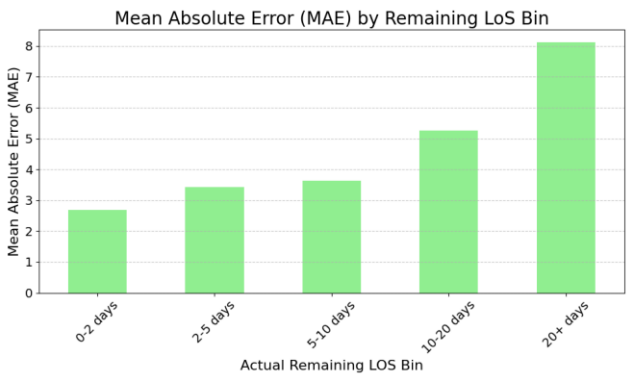

Figure 11: Prediction bins

Therefore, the model lacks useful features in predicting long stay patients in the ICU such as coma patients.

2) Feature importance

SHAP was used to show the feature importance:

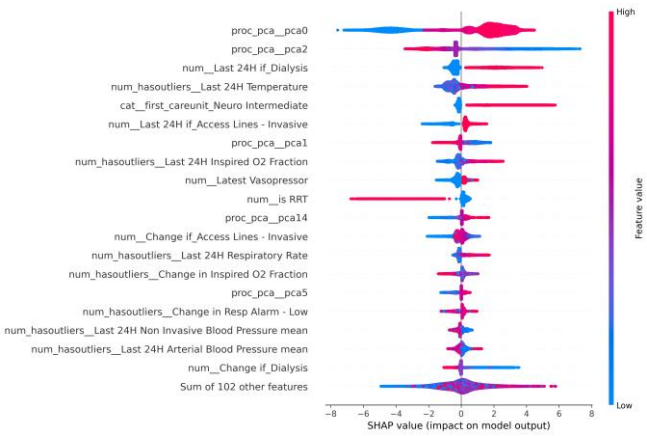

Figure 12: SHAP values

The top 2 features are related to embeddings of the concatenated list of procedures that the patient has taken in the ICU to date. Since the original feature was projected to 50 dimensions in PCA, it's challenging to find more insights into the importance of the text. The approach taken was to list common embeddings with the highest score for PCA 0 and 2 components. Since PCA 2's score was inversely related to importance (as shown in the color), its score is multiplied by -1. The list of procedures that was encoded as the top 2 embeddings included the following texts from 2 patients:

"Insertion of Infusion Device into Right Atrium, Percutaneous Approach, Insertion of Infusion Device into Superior Vena Cava, Percutaneous Approach, Ultrasonography of Heart.."

"Interruption of the vena cava, Temporary tracheostomy, Other excision of joint, other specified sites, Other cervical fusion of the anterior column, anterior technique, Enteral infusion of concentrated nutritional substances, Closed [endoscopic] biopsy of bronchus"

In order to understand common keywords, the top 15 embeddings from both PCA0 and PCA2 were combined and stop words applied before generating a word cloud of the frequent keywords:

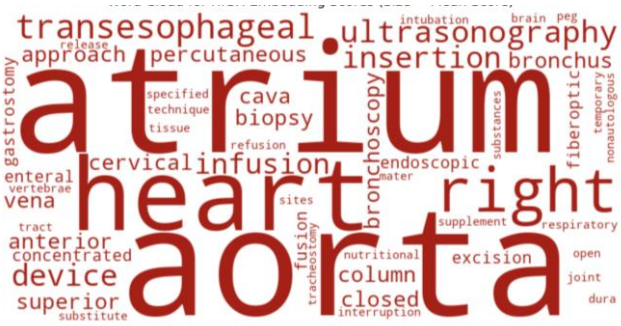

Figure 13: Word cloud of frequent terms in list of procedures

This showed that heart-related procedures tend to lengthen the patient stay in ICU for this hospital.

The 3rd most important feature is whether a patient has undergone dialysis in the last 24 hours followed by whether the temperature was rising in the last 24 hours, and if the first care unit was neuro intermediate – this is the unit that takes care of patient with neurological conditions such as strokes, seizures and brain related injuries.

B. Admission Forecasting

1) Model Performance across Different Horizons

All the candidate models as well as baseline models are fit and tested across different forecast horizons ($d = 3, 5, 7$ days). All the models are tuned and fit based on their own properties.

The results shown in Table 1. revealed that a 30-day moving average model consistently outperforms the other models no matter for single-point forecasts or aggregate totals, followed by two exponential smoothing models – Holt-winters seasonal or Holt's Linear model.

Among all horizons, the best-performing model moving average achieved the lowest MAPE (14.79%) when $d = 7$ and the RMSE for single-point forecasts reached the lowest at 2.66 as $d = 5$.

	Models	Single-Point		Aggregated forecasts		
		MAE	RMSE	MAE	RMSE	MAPE(%)
d=3	MA (30 days)	**2.66**	**3.46**	**4.92**	**6.35**	-
	Holt-Winters Seasonal	2.71	3.52	5.11	6.61	-
	Holt linear	2.71	3.52	5.10	6.60	-
	ARIMA	2.81	3.58	5.46	6.93	-
d=5	MA (30 days)	**2.66**	**3.46**	**6.77**	**8.70**	**16.92**
	Holt-Winters Seasonal	2.71	3.51	7.04	9.22	17.73
	Holt linear	2.71	3.51	7.06	9.23	17.78
	ARIMA	2.81	3.59	7.66	9.88	23.93
d=7	MA (30 days)	**2.68**	**3.48**	**8.47**	**11.03**	**14.79**
	Holt-Winters Seasonal	2.73	3.54	8.86	11.84	15.88
	Holt linear	2.73	3.53	8.90	12.57	15.91
	ARIMA	2.84	3.61	9.70	12.86	21.90

Table 1. Evaluation metrics of admission forecasts (test set)

2) Residual Diagnostics

Ljung-Box test is a widely-used statistics test to measure the performance of time series model by checking the independence of forecast residuals. The null hypothesis of it is that the residuals are independently distributed under specified lags, typically indicating a better model fit.

Given that the overlapping windows in rolling-origin forecasts would inherently introduce systematic dependencies, we resampled the residuals with stride d prior to the residual diagnostics, so as to eliminate such effect.

The results are shown in Table 2, listing the Q statistics and corresponding p-values for tests spanning lags order 1 to 10. Some lags still exhibited significant auto-correlation at 0.01 significance level, which means the model are not yet perfect fit.

Lags	Q statistic	p-value
1	6.07	0.0137
2	6.11	0.0471
3	13.06	**0.0045**
4	16.60	**0.0023**
5	17.76	**0.0033**
6	17.84	**0.0067**
7	18.31	0.0106
8	18.39	0.0185
9	18.41	0.0307
10	18.99	0.0404

Table 2. Ljung-box Test of Moving Average (30 days) as $d = 7$.

These results are also aligned with ACF plots. A certain degree of short-term auto-correlation persists even though there is no obvious skewness or abnormalities in the residuals. However, the auto-correlation is relatively weak and basically acceptable.

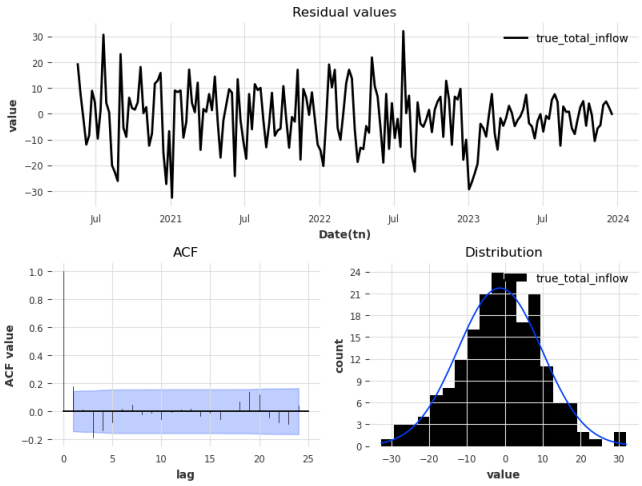

Figure 3. Residual diagnostics of 30-day MA as $d = 7$

3) Error analysis

Even though the evaluation metrics show that the moving average models are consistently the best among all, the results are still not fully satisfactory. Given the simplicity of moving average models and the results originated from STL decomposition, we found it necessary to further research in the reasons.

By examining the predicted and actual admission curves, it's clearly revealed that the exponential models are less responsive to the sharp decline in 2023, as they inherently smooth out the fluctuations while assuming the long-term stability of the time series. Moving average models, however, only have short-term memory, making it more sensitive to the sudden changes.

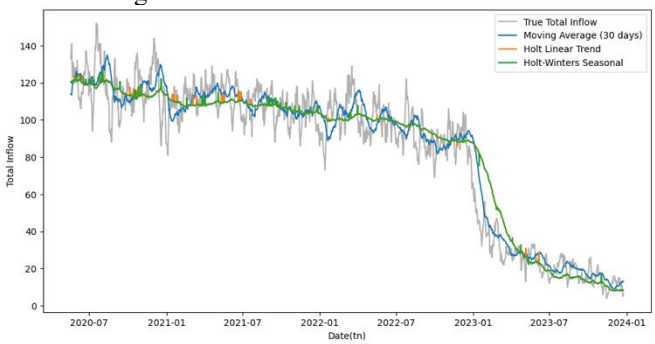

Figure 4. True total admissions v.s predicted admissions

If excluding the data after 2023, the evaluations result are completely reversed. Holt's linear trend models and Holt-winter seasonal models achieve higher predictive accuracy with lower MAE and RMSE, aligning more closely with theoretical expectations.

Models	Single-Point		Aggregated forecasts		
	MAE	RMSE	MAE	RMSE	MAPE (%)
MA (30 days)	3.04	3.85	9.30	11.72	9.01
Holt-Winters Seasonal	**3.01**	3.79	**8.62**	**10.80**	**8.31**
Holt linear	3.01	**3.79**	8.66	10.81	8.34

Table 3. Metrics of admission forecasts (d=7, before 2023t)

Same phenomenon occurred for Ljung-Box test, the residual auto-correlation significantly decreased not only for holt-winter seasonal models but also moving average models.

Lags	30-day MA		Holt Winter Seasonal	
	Q statistic	p-value	Q statistic	p-value
1	1.31	0.2529	2.04	0.1531
2	1.96	0.3748	2.60	0.2729
3	12.30	**0.0064**	3.71	0.2950
4	16.82	**0.0021**	3.81	0.4322
5	17.95	**0.0030**	4.96	0.4204
6	17.99	**0.0062**	5.20	0.5186
7	18.51	**0.0099**	5.20	0.6357
8	18.51	0.0177	5.75	0.6753
9	18.73	0.0276	6.49	0.6905
10	19.23	0.0374	7.39	0.6885

Table 4. Ljung-box Test results of Moving Average (30 days) and Holt's Winter Seasonal model as $d = 7$.

C. Hybrid

1. Hybrid Model Performance Overview

The hybrid model integrates forecasts from the admission model (new inflow) and the LoS model (expected discharges), generating short-term ICU occupancy predictions for 3-, 5-, and 7-day horizons. After completing the test-set imputation and aligning the reassigned timeline (2020/5/19 - 2023/12/31), we evaluated both absolute occupancy levels and occupancy changes, supported by the corresponding visualizations.

2. Occupancy Prediction Results

Performance Metrics

horizon	n	MAE	MSE	RMSE	MAPE_%
3-day	1319	22.47	652.16	25.54	35.64
5-day	1317	33.03	1427.74	37.79	53.13
7-day	1315	45.19	2441.22	49.41	73.16

Across all horizons, the model captures the overall trend of ICU occupancy; however, errors increase with forecast horizon, which is expected in cumulative multi-step forecasting. The positive bias at every horizon indicates a tendency for systematic overestimation. This pattern is confirmed visually: scatter plots of residuals show predominantly positive errors, and the time series overlays reveal the predicted lines sitting above the ground-truth occupancy.

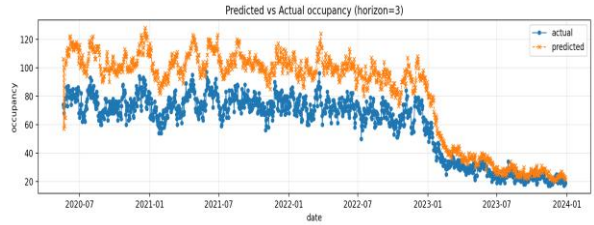

Figure 5. Predicted vs. Actual Occupancy (h=3)

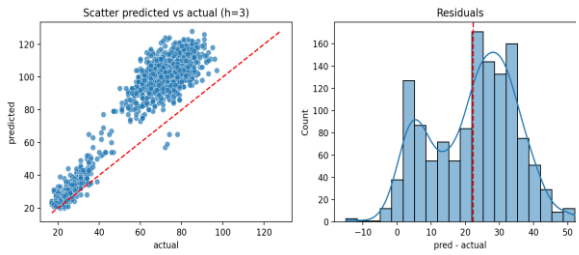

Figure 6. Residual Plot for Occupancy Predictions(h=3)

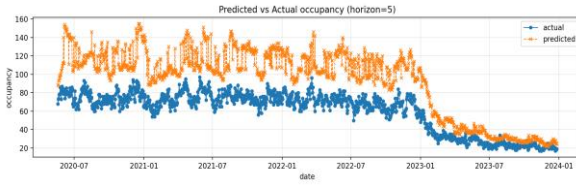

Figure 7. Predicted vs. Actual Occupancy(h=5)

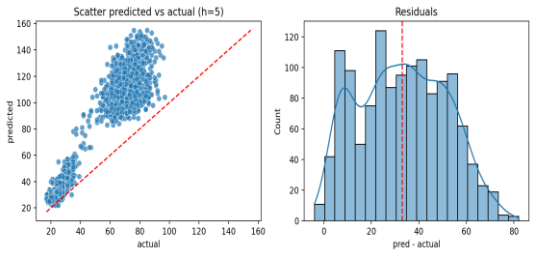

Figure 8. Residual Plot for Occupancy Predictions(h=5)

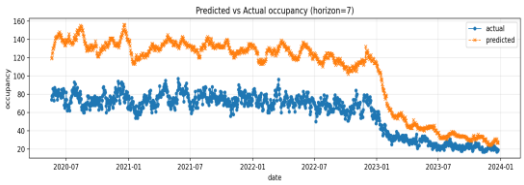

Figure 9. Predicted vs. Actual Occupancy(h=7)

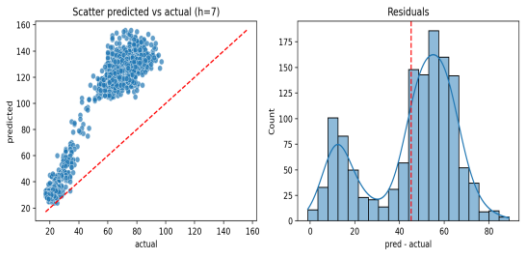

Figure 10. Residual Plot for Occupancy Predictions(h=7)

Interpretation

- Short-term accuracy (3-day horizon) is reasonable: MAE of ~22 beds correspond to less than one ICU ward's average capacity.
- Increasing horizon → increasing compounding error, mainly driven by:
    - accumulating admission prediction uncertainty,
    - imperfect discharge estimation for existing patients,
    - missing discharge events for short-stay newly admitted patients.
- Bias is the dominant source of error, not random fluctuation; this implies a structural component in the model rather than noise

3. Occupancy-Change Prediction Results

Performance Metrics and Visualizations

horizon	n	MAE	MSE	RMSE
3-day	1316	7.23	91.61	9.57
5-day	1312	10.89	213.52	14.61
7-day	1308	7.28	88.81	9.42

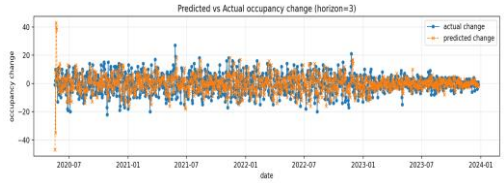

Figure 11. Predicted vs Actual Occupancy Change (h=3)

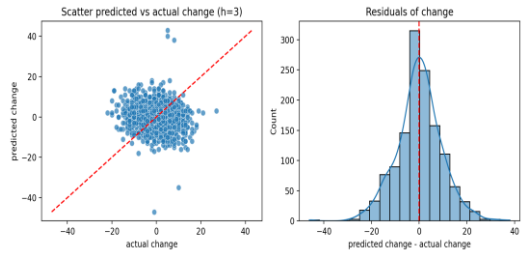

Figure 12. Residual Plot for Occupancy Change Predictions (h=3)

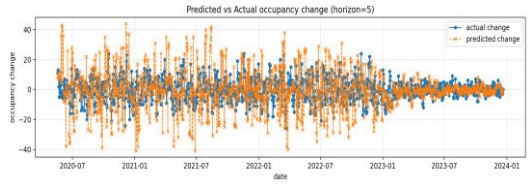

Figure 13. Predicted vs Actual Occupancy Change (h=5)

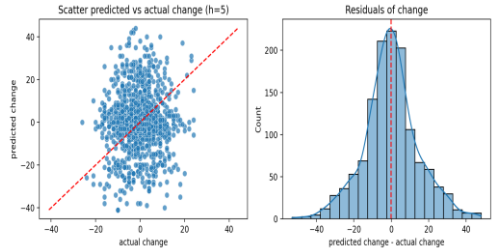

Figure 14. Residual Plot for Occupancy Change Predictions (h=5)

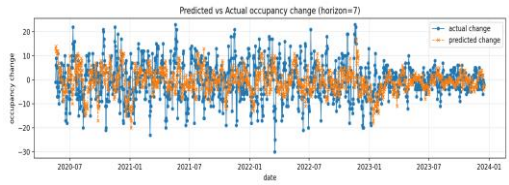

Figure 15. Predicted vs Actual Occupancy Change (h=7)

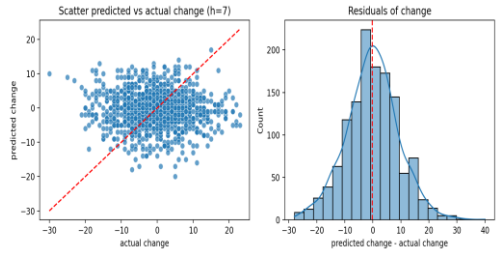

Figure 16. Residual Plot for Occupancy Change Predictions(h=7)

Interpretation

- For horizons 3 and 7, RMSE stabilizes around 9–10 beds, suggesting consistent ability to track short-term increases or decreases.
- The 5-day horizon is the most volatile, potentially due to mismatch between admission uncertainty and discharge uncertainty at that temporal distance.
- Visualizations show that predicted changes correctly identify many upward/downward movements even when the magnitude deviates.

4. Analytical Insights and Observations

Insight 1: Structural Overestimation Rooted in Short-Stay Patients

The dominant error source is upward bias. This is attributable to a structural limitation:

- The current hybrid framework only estimates outflow from existing patients.
- It does not estimate the LoS for newly predicted admissions, so patients who are admitted and discharged within the forecast window are not counted as discharges.

These "short-stay" patients appear frequently in real ICU operations but are systematically omitted, creating consistent positive bias in occupancy.

Insight 2: Occupancy Change is a More Stable Target

Despite level-bias issues, occupancy-change predictions remained relatively stable across horizons. This indicates the model captures dynamic behavior better than absolute magnitude, implying:

- the models understand the temporal patterns of inflow and outflow,
- but cumulative summation of errors in level forecasts magnifies small deviations into significant absolute bias.

This insight motivates future modeling that places greater emphasis on incremental changes rather than absolute counts.

5. Unexpected Findings

Several unexpected patterns emerged from the hybrid model evaluation, offering deeper insights into its behavior beyond what the numerical metric alone revealed.

First, the residual visualizations displayed clear horizon-dependent structures: while the 3-day horizon residuals were relatively compact, the 7-day horizon showed wider dispersion and banded patterns. This suggests that forecast errors accumulate in a non-linear fashion as the horizon extends, reflecting differential sensitivity to admission versus discharge uncertainty over time.

Second, the 7-day occupancy-change performance was unexpectedly comparable to that of the 3-day horizon. Longer horizons typically deteriorate due to compounding uncertainties, yet the similar error magnitude suggests that weekly-scale patterns in admissions and discharges may exert a smoothing effect that stabilizes longer-horizon change predictions.

VII. Discussion of Challenges & Lessons Learned

A. *Key Challenges*

Challenge 1: Reconstructing ICU Temporal Dynamics Due to Year-Shifted Admissions

Because admission years in the original dataset were shifted for privacy protection, the true temporal sequence of ICU events was unknown. Only the year group of each record was available, making it impossible to accurately reconstruct actual occupancy dynamics or validate temporal trends.

To approximate real-world patterns, we applied a random year reassignment within each year group, addressed leap-year inconsistencies by mapping February 29 to February 28 when necessary, and preserved each patient's original LoS. This approach preserved daily structure more faithfully than aggregated statistics or fixed-year assignments.

Challenge 2: Processing and Managing Large-Scale Clinical Data

The chartevents table contains more than 313 million rows, posing substantial computational and storage challenges. Loading the full table directly was time-consuming and made routine tasks—such as feature selection, merging, and consistency checks—practically infeasible.

To manage this, the dataset was processed in smaller chunks, allowing incremental loading, cleaning, and saving. A preliminary diagnostic review was first conducted using a single representative chunk to identify key variables and test all processing steps before scaling up to the full dataset. This workflow improved computational feasibility and stability but required significant engineering effort and careful planning to avoid data corruption or inconsistencies.

Challenge 3: Integrating Admission and LoS Models into a Unified Hybrid Framework

A major methodological challenge involves combining two models that operate at different granularities: the admission model predicts daily inflow, while the LoS model predicts patient-day–level remaining length of stay. Building a hybrid occupancy prediction framework required aligning their timelines, standardizing date references, and constructing a complete patient-level test set that could support hybrid computation.

This integration was further complicated by irregularities in the LoS predictions, including missing records, duplicated predictions for the same patient-day, and inconsistent LoS trajectories. To ensure a usable dataset, we implemented a multi-step imputation strategy: averaging duplicated predictions, inferring missing values based on intra-stay temporal position, and using daily average LoS when no patient-level information was available. Although this allowed hybrid model evaluation, it introduced additional uncertainty and highlighted the inherent complexity of integrating multi-component forecasting models.

B. *Lessons Learned*

This project offered several key lessons across both modeling and workflow design. First, it highlighted the critical importance of maintaining strict alignment across model components in a hybrid system. Even minor inconsistencies in time indexing or data structure can propagate through downstream calculations, underscoring the need for rigorous preprocessing and validation. Second, the

project demonstrated the value of systematic diagnostic analysis. Residual inspections, horizon-specific evaluations, and visualization-based checks revealed structural issues that were not evident from aggregate metrics alone, reinforcing the need to pair quantitative evaluation with qualitative assessment. Third, the contrasting performance between occupancy-level and occupancy-change predictions emphasized the importance of selecting targets aligned with operational needs. More stable and informative targets can significantly improve interpretability and robustness. Finally, the iterative troubleshooting and cross-team collaboration throughout the project strengthened project management, communication, and practical problem-solving skills—capabilities essential for real-world data science work.

VIII. Conclusion and Future Work

A. Conclusion

This project developed a hybrid ICU occupancy forecasting framework that jointly models admission inflow and patient-level remaining length of stay (LoS), addressing the limitations of existing approaches that relay on naive or averaged LoS assumptions. By integrating a rolling-origin admission prediction model with a granular, individual-based LoS model, the proposed system provides a more realistic representation of ICU dynamics and supports short-term operational decision-making.

Our results show that the hybrid model effectively captures overall occupancy trends across 3-, 5-, and 7-day horizons. Although the model exhibits systematic overestimation – primarily due to unmodelled short-stay admissions that both enter and exit within the prediction window- it consistently tracks directional changes in occupancy. The stability of occupancy-change predictions across horizons highlights the framework's ability to detect short-term increases or decreases in ICU demand, which is crucial for advance planning.

The project successfully addressed the problem statement by explicitly modelling both inflow and outflow processes, thereby overcoming the underestimation issues associated with average-based LoS assumptions. It met the stated objectives by improving the interpretability and responsiveness of bed occupancy forecasts and by demonstrating how individual-level LoS predictions can be incorporated into daily operational planning. The LoS showed higher accuracy for short term patients vs long term patients in ICU. As such, additional features such as SOFA[20] (Sequential Organ Failure Assessment) could be computed to detect coma patients.

Overall, the hybrid model represents a meaningful step toward data-driven ICU capacity management. Its ability to provide offers practical value for hospital administrators, and its structure lays out the groundwork for future extensions that incorporate newly admitted patients' LoS, probabilistic discharges, or real-time clinical data. The framework reinforces the importance of integrating heterogeneous predictive components to enhance the reliability of short-term ICU bed forecasting.

B. Recommendations for Future Work

Several directions can further strengthen and extend the current hybrid modeling framework. First, the discharge component could be enhanced by modeling expected LoS for newly admitted patients, thereby capturing short-stay discharge events that the current system structurally overlooks. This would reduce the persistent upward bias in occupancy forecasts and improve consistency across horizons.

Second, future work could explore probabilistic or survival-based discharge models that explicitly account for uncertainty in LoS. Such approaches may provide more realistic discharge distributions and mitigate compounding error as the forecast horizon increases.

Third, the interaction between the admission and LoS models could be improved by developing joint or end-to-end architectures, ensuring that inflow and outflow predictions remain internally consistent. This may reduce erratic behavior observed at intermediate horizons, such as the unexpectedly unstable 5-day results.

Finally, given that occupancy-change predictions exhibited greater robustness than absolute occupancy levels, future efforts could prioritize modeling incremental changes or employing hybrid formulations that directly target net inflow. Such reframing may produce more stable and operationally meaningful forecasts, especially in dynamic ICU environments.

IX. Group Member Contributions

Provide a clear and detailed account of each group member's specific contributions to the project. Outline the roles and responsibilities handled by each person, including tasks completed and areas of focus (sample shown in Table 1). This transparency demonstrates teamwork and individual efforts.

Table 1. Summary of Individual Contributions

Member	Roles and Key Contributions
Chen Yu	Conducted data merging from the raw MIMIC-IV tables to create the modeling dataset, including feature selection, data transformation and aggregation. Developed and evaluated admission forecasting models using time series techniques such as moving averages, exponential smoothing, and ARIMA. Performed residual diagnostics and error analysis to assess model fit and identify areas for improvement. Led the team coordination and checked the project progress.
Zhang Ziyi	Conducted multivariate exploratory data analysis, including relationships between categorical–categorical, numerical–numerical, and numerical–categorical variables. Designed the hybrid model framework，including unifying the timelines and outputs of the admission and LoS models and performing test set imputation to ensure temporal consistency, complete patient-day coverage, and aligned inputs for hybrid evaluation. Evaluated hybrid model performance across multiple horizons using occupancy-level and occupancy-change metrics, including error analysis and visualizations.

Ong Gim Shin	Data preprocessing which includes data cleaning, transformation, missing values and outliers handling with Python. Feature engineering which includes aggregation of time series data, generating new features from other tables such as input events, tf idf and text embeddings from procedure names (and PCA). Training and hyperparameter tuning of models for LoS predictions which includes building the sklearn pipeline for preprocessing different types of features. Feature importance and explainability of LoS predictions.
Yu Yichen	Exploratory data analysis, including checking data quality, examining variable distributions, and exploring patterns and trends in the ICU data. Exploratory data analysis report generation, including using Python code to produce summary plots and tables used by the team for visualizing and summarizing key data patterns. Dataset reviewing, including checking intermediate datasets for consistency. Report writing, including drafting assigned sections based on the project structure. General project support, including assisting with coordination tasks and contributing to miscellaneous deliverables as required.

X. List of References

References

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APPENDIX (NOT SUBJECT TO PAGE LIMIT)

Include supplementary details such as an extended description of dataset variables, larger versions of tables/figures, additional data visualizations, or any other supporting material. The appendix is not subject to the 12-page limit.

A. Extended Description of Dataset Variables

Provide a comprehensive description of the dataset variables, including detailed explanations, units of measurement, and any relevant coding schemes used.

1. **Hospital admission data**:

Feature	Meaning	Data Type
gender	The patient's gender (e.g., 'M', 'F')	Categorical (Binary)
age	The patient's age at the time of ICU admission	Numerical (Integer)
admission_type	The category of the patient's hospital admission (e.g., 'EMERGENCY', 'ELECTIVE', 'URGENT')	Categorical
insurance	The patient's insurance type (e.g., 'Medicare', 'Private', 'Medicaid')	Categorical
marital_status	*The patient's marital status (e.g., 'MARRIED', 'SINGLE', 'WIDOWED')*	*Categorical*

Table 1: Features available upon hospital admission

2. **ICU admission data**:

Feature	Meaning	Data Type
first_careunit	*The first intensive care unit the patient was admitted to during their hospital stay (e.g., 'Medical ICU (MICU)', 'Coronary Care Unit (CCU)', 'Surgical ICU (SICU)')*	*Categorical*

Table 2: Features available upon ICU admission

3. **Chart events (Time Series)**:

Feature	Meaning	Data Type
ART BP Mean / Arterial Blood Pressure mean (Invasive)	The average blood pressure measured via an invasive arterial line	Numerical (float)
Non Invasive Blood Pressure mean (NIBP)	The average blood pressure measured with a standard arm cuff	Numerical (float)
Heart Rate	The patient's heart rate in beats per minute (bpm)	Numerical (float)
Respiratory Rate	The patient's number of breaths per minute	Numerical (float)
Temperature	The patient's core body temperature	Numerical (float)
Inspired O2 Fraction (FiO2)	The concentration of oxygen the patient is breathing (room air is 21%). High values mean the patient requires significant oxygen support	Numerical (float)
Weight	The patient's daily weight. In the ICU, this is a proxy for fluid balance	Numerical (float)
Heart rate alarm - high	Alarm Settings (Proxy for Clinical Judgment) for heart rate. These features represent the low alarm limit set by the clinical staff on the patient's bedside monitor	Numerical (integer)
Resp alarm - high	Alarm Settings (Proxy for Clinical Judgment) for respiratory rate. These features represent the high alarm limit set by the clinical staff on the patient's bedside monitor	Numerical (integer)
Resp alarm - low	Alarm Settings (Proxy for Clinical Judgment) for respiratory rate. These features represent the low alarm limit set by the clinical staff on the patient's bedside monitor	Numerical (integer)
Vasopressor	Vasopressor use indicates hemodynamic instability (shock), i.e. Epinephrine, Dobutamine, Dopamine, Phenylephrine, Norepinephrine, Vasopressin	Numerical (integer)
if_Access Lines - Invasive	Invasive lines - medical devices, such as catheters and cannulas, inserted into the body to provide access to the bloodstream or other internal systems for administering fluids, medications, or for monitoring	Categorical (Binary)
if_Cardiovascular	Cardiovascular support	Categorical (Binary)
if_Dialysis	Dialysis support	Categorical (Binary)
if_IABP	Intra-Aortic Balloon Pump are mechanical heart pumps, indicating extreme heart failure	Categorical (Binary)
if_Impella	Ventricular Assist Device are mechanical heart pumps, indicating extreme heart failure	Categorical (Binary)

High risk (>51) interventions	High-risk interventions based on the Therapeutic Intervention Scoring System[3]	Categorical (Binary)
Low risk (25-50) interventions	Low-risk interventions based on the Therapeutic Intervention Scoring System[3]	Categorical (Binary)

Table 3: Time-series measurements available throughout ICU stay

4. **Procedures**:

Feature	Meaning	Data Type
Procedures	Procedures taken by the patient while in ICU with associated start and end times	Text

Table 4: Names of procedures taken available throughout ICU stay

B. Larger Tables and Figures

Include larger versions of tables or figures here if they were scaled down in the main text for readability. Ensure all figures and tables are clearly labeled and referenced.

C. Additional Data Visualizations

Present any additional visualizations that support the findings discussed in the main report. This may include charts, graphs, heatmaps, or other graphical representations.

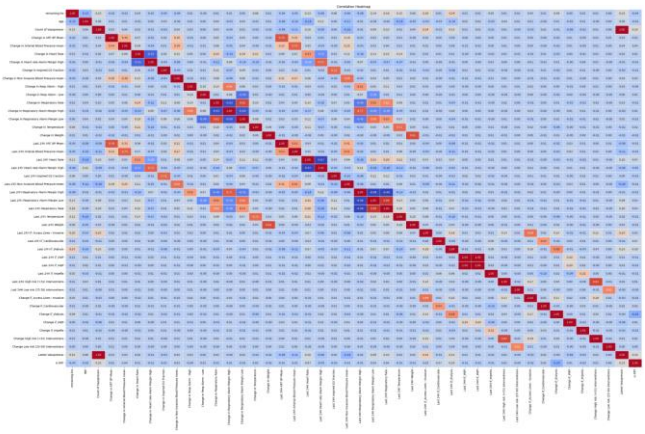
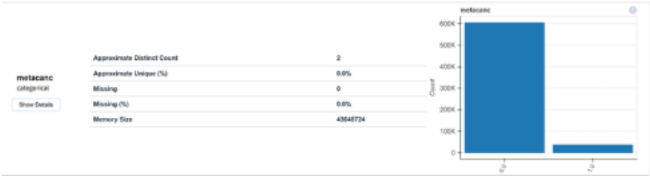


Fig.A1: Heatmap

Fig.A2: Descriptive Statistics and Distribution of aids

Fig.A3: Descriptive Statistics and Distribution of dementia



eta Fig.A4: Descriptive Statistics and Distribution of metacanc

Fig.A5: Descriptive Statistics and Distribution of stay_days

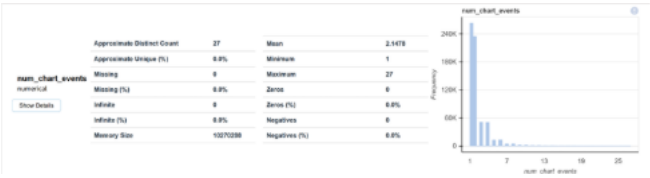


Fig.A6: Descriptive Statistics and Distribution of num_chart_events

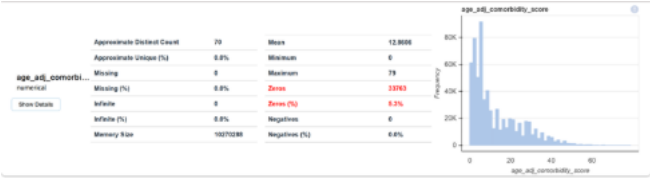


Fig.A7: Descriptive Statistics and Distribution of anchor_age